SeedSigner: rolling a seed in dice mode

99 rolls → 24 words · menu path SeedSigner OS 0.8.7 · independent verification · Self-Experiment 1b · belongs with dice log sheet 1b

⚠ **The string of digits IS the seed.** SHA-256 turns the 99 digits into the 24 words. Whoever knows the digits knows the wallet. The filled-in dice log sheet is as secret as the words.

⚠ **Always treat the first run as practice** with a throwaway seed that never holds real coins afterwards. Only once menu, sequence and verification are familiar do you roll the productive seed. ⚠ **No camera in the room:** phone away, webcam unplugged, mind windows and mirrors.

PREPARATION

- ☐ Dice log sheet 1b is ready (20 rounds of 5 digits = 99, last round 4)
- ☐ Dice setup: cup with a lid (an ice cream tub does the job), throwing surface with a rim (box lid), five dice of the same kind, casino quality
- ☐ SeedSigner on a power bank or mains adapter, around 45 seconds to boot. **Then pull the microSD**: the device runs entirely from RAM and says so itself: "You can remove the SD card now". With no card in it, nothing can be written to it. Permanent: Settings → Persistent settings → Enabled, then Advanced → MicroSD notification duration → Until SD removed.
- ☐ Paper for the fingerprint (8 characters) and for device labels that differ, a clock

MENU PATH ON THE DEVICE (SEEDSIGNER OS 0.8.7)

1 ☐ Main menu → **Tools**

2 ☐ There are **two** entries called "New seed": the first with a **camera icon** (image entropy), the second with a **dice icon**. **Choose the second one.** Easy to mix up: the camera variant is legitimate too, but it cannot be recomputed.

3 ☐ Screen "Mnemonic Length": choose **"24 words (99 rolls)"**, not "12 words (50 rolls)".

4 ☐ Screen **"Dice Roll 1/99"**: 3×3 pad with dice pips 1 to 6. For each roll select the number and press, the title counts up (2/99, 3/99 …). Mistyped? The delete key (⌫, bottom right of the pad) takes back the last digit, then reconcile counter and sheet again.

5 ☐ After the 99th digit the device moves on by itself, no confirmation needed.

6 ☐ Warning screen **"Caution – Classified Info!"** (keep seed words private, away from online devices) → **"I understand"**

7 ☐ **"Seed Words 1/6" through "6/6"**, four words per page, "Next" until "Done". With the practice seed write the words down for comparison. With the real seed only by hand, as the template for the steel (chapter 5).

8 ☐ Prompt **"Verify" / "Skip"** → **Verify**: the device asks for all 24 words one at a time, four options each. Play along, this is the first backup test.

9 ☐ The final screen shows the **fingerprint** (8 characters): _____ note it down → "Done". If "BIP-39 Passphrase" appears: do not use it for multisig cosigners.

10 ☐ The seed now lives in RAM only (menu "Seeds"). Power gone = seed gone. For the practice run that is the point.

How the rolling goes: Per round shake five dice, throw, write the digits onto the sheet in a **fixed reading order** (left to right), then tap those same five into the device, **reconcile the device counter with the sheet.** That check every five rolls is the error brake. No roll is repeated because the number is not to your liking. Dice that land tilted or stacked: repeat the whole round, note it on the sheet.

Why 99 rolls: Each roll carries $\log_2 6 \approx 2.58$ bits, 99 rolls around 256 bits, as much as 24 words hold (50 rolls = 12 words). The SeedSigner takes the 99 digits as text, forms SHA-256 over them and derives the words per BIP39. That is exactly why it can be recomputed.

VERIFICATION: NOT HAVING TO BELIEVE THE DEVICE

A ☐ **Your own check script** (in the workshop package: `verify_dice_seed.py`, Python standard library only plus the official BIP39 word list, no SeedSigner code): start `python3 verify_dice_seed.py`, type the 99 digits when asked (that keeps them out of the command history) → compare the 24 words one by one with the display. Run `--selftest` first (checks against the examples in the SeedSigner documentation). The script sends and stores nothing. Without a terminal, use the Dice Check in the browser (`dice-check.html`), and all ways are on the Dice Check guide.

B ☐ **Second code base** (optional): Ian Coleman's `bip39-standalone.html` offline → "Show entropy details" → actively click **"Base 10"** (**never "Dice [1-6]"**: it turns a 6 into a 0) → set **"Mnemonic Length" to "24 Words"** (otherwise "Use Raw Entropy" stays active and nothing is hashed) → enter the digits → compare.

⚠ ☐ **What is tested is the method, not the real seed:** if the device computes practice digits correctly, the procedure is established. For the real seed that is enough in practice, but it cannot be proven (chapter 3). If you do recompute it anyway: **offline only** (Tails without network), then shut down. Terminal, history and clipboard see everything.

TIME

Start: _____

Rolling done: _____

Verification done: _____

Date: _____

SNAGS / LABELS THAT DIFFERED (FOR THE LOGBOOK)

_____

_____

_____

RESULT

- ☐ **Passed:** The 24 words from your own computation are identical with the display. The SeedSigner computed correctly, and the seed does not hang on a manufacturer's randomness.
- ☐ **Mismatch:** Discard the practice seed, check the digits on the sheet against what you entered (most common cause: a typo between sheet and device), repeat the run. A real seed is only rolled once the practice run goes smoothly.

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